1 install and deploy

Base: github, docker, docker compose

Github: https://github.com/COMP6016-25S2-G2-AIclinic-ECG/orthanc

After Glone the project, we can see 3 scripts for different os.

Available Scripts

Windows Batch Script

- **File:** `run_all_docker_compose_windows.bat`
- **Usage:** Run this script in Command Prompt or PowerShell to start all AIClinic services in the correct order.
- **Command:**

```
.\run_all_docker_compose_windows.bat
```

Windows PowerShell Script

- **File:** `run_all_docker_compose_windows.ps1`
- **Usage:** Run this script in PowerShell to start all AIClinic services in the correct order.
- **Command:**

```
.\run_all_docker_compose_windows.ps1
```

Linux Bash Script

- **File:** `run_all_docker_compose_linux.sh`
- **Usage:** Run this script in a Linux terminal to start all AIClinic services in the correct order.
- **Command:**

```
# Make the script executable first (if not already)
chmod +x run_all_docker_compose_linux.sh

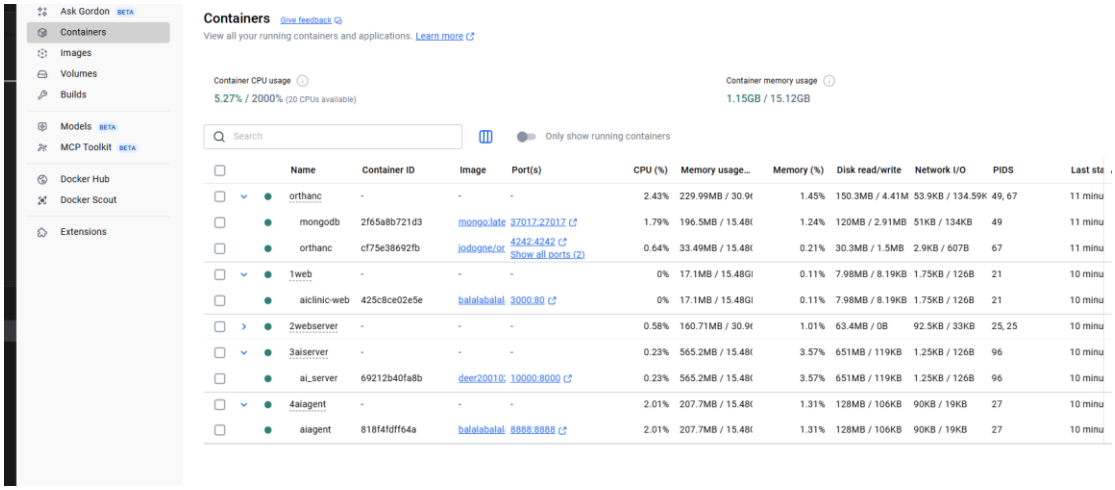
# Then run the script
./run_all_docker_compose_linux.sh
```

Now we choose the right script for our own os and run it.

```
PS D:\study\csproject\AIClinic\orthanc\orthanc> .\run_all_docker_compose_windows.bat
Starting all AIClinic docker services...
Step 1: Starting Orthanc and MongoDB services...
[+] Running 2/2
 ✓ Container orthanc  Running                                                0.0s
 ✓ Container mongodb  Running                                                0.0s
Orthanc and MongoDB services started successfully.
Waiting for Orthanc to be ready...
Orthanc is ready
Step 2: Starting Web Frontend...
time="2025-09-25T11:01:06+08:00" level=warning msg="D:\\study\\csproject\\AIClinic\\orthanc\\orthanc\\1 web\\docker-compose.yml:
lete, it will be ignored, please remove it to avoid potential confusion"
[+] Running 2/2
 ✓ Network 1web_default    Created                                           0.1s
 ✓ Container aiclinic-web  Started                                           0.8s
Web Frontend started successfully.
Step 3: Starting Web Server...
time="2025-09-25T11:01:07+08:00" level=warning msg="D:\\study\\csproject\\AIClinic\\orthanc\\orthanc\\2 web server\\docker-compo
is obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] Running 2/2
 ✓ Container webserver      Started                                          1.0s
 ✓ Container webserverauth  Started                                          0.9s
Web Server started successfully.
Step 4: Starting AI Server...
```

```
Step 5: Starting AI Agent...
time="2025-09-25T11:01:10+08:00" level=warning msg="D:\\study\\csproject\\AIClinic\\orthanc\\orthanc\\4 ai
time="2025-09-25T11:01:10+08:00" level=warning msg="D:\\study\\csproject\\AIClinic\\orthanc\\orthanc\\4 ai
obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] Running 1/1
obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] Running 1/1
[+] Running 1/1
 ✓ Container aiagent  Started                                    0.9s
 ✓ Container aiagent  Started                                    0.9s
AI Agent started successfully.
All AIClinic services have been started successfully
You can access the following services:
- Orthanc Web Interface: http://localhost:8042
- AIClinic Web Frontend: http://localhost:3000
- Web Server API: http://localhost:8000
- Auth Server: http://localhost:8001
- AI Server: http://localhost:10000
- AI Agent: http://localhost:8888
```

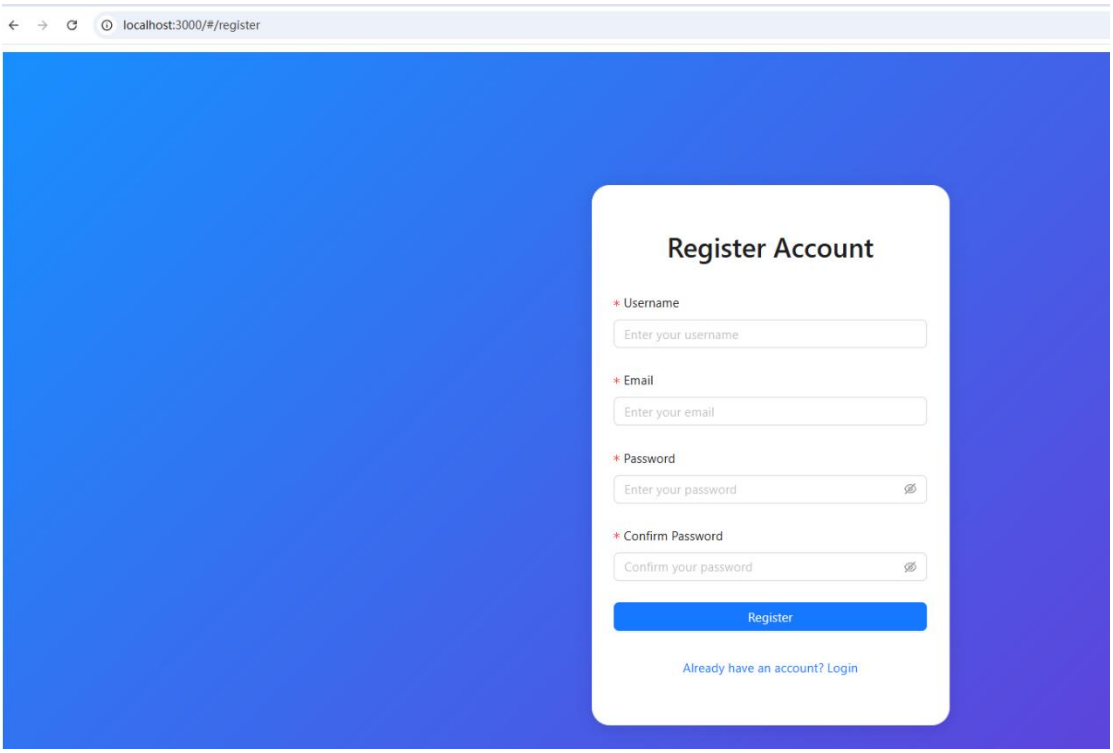
After AIClinic services have been started successfully. We could start next step.



2 how to use our platform

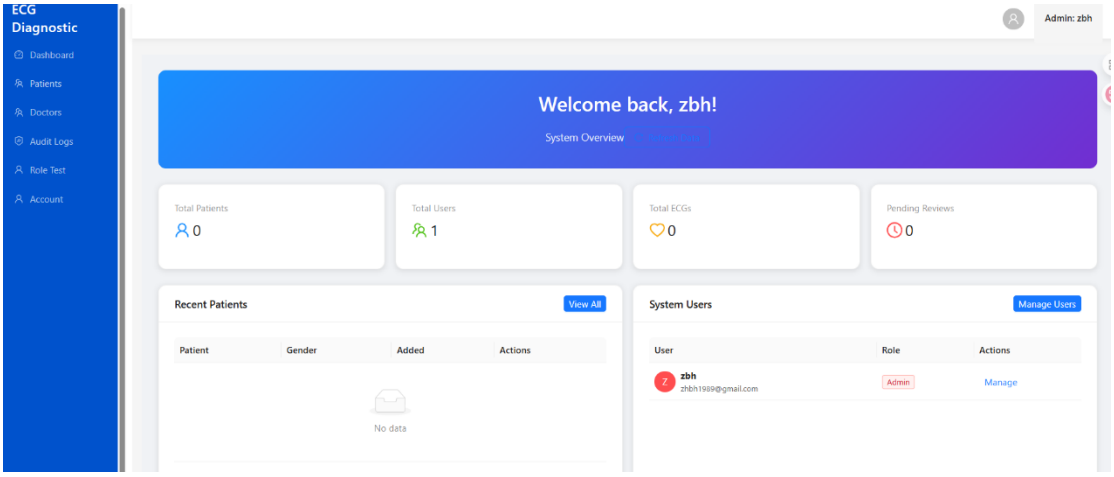
Step 1 register

http://localhost:3000/#/register



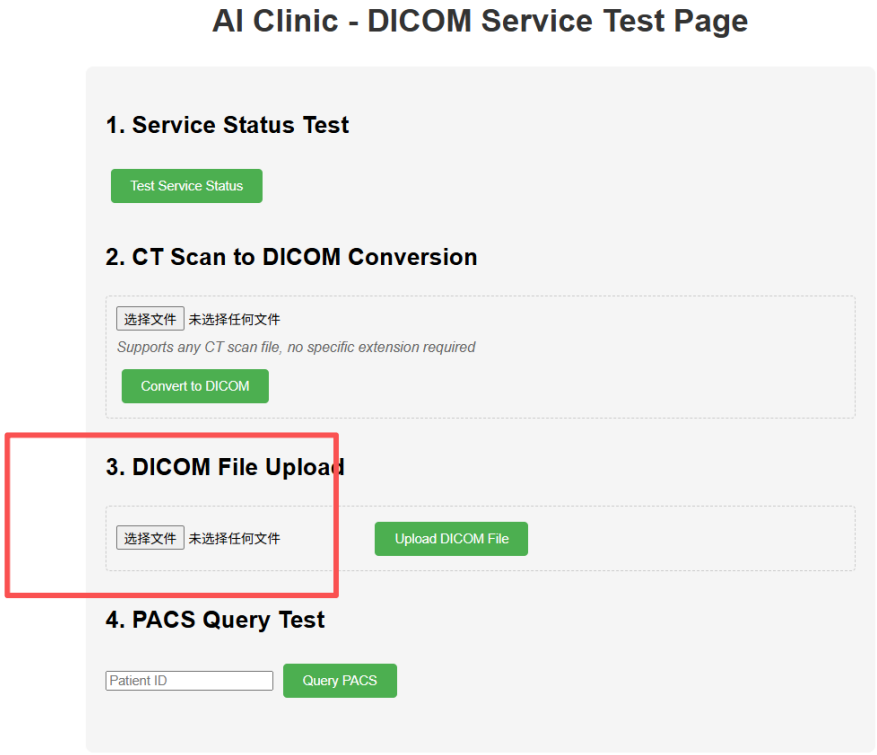
After we register our account and login successfully, we can see the database and orthanc server is empty now.

Step 2 create a patient

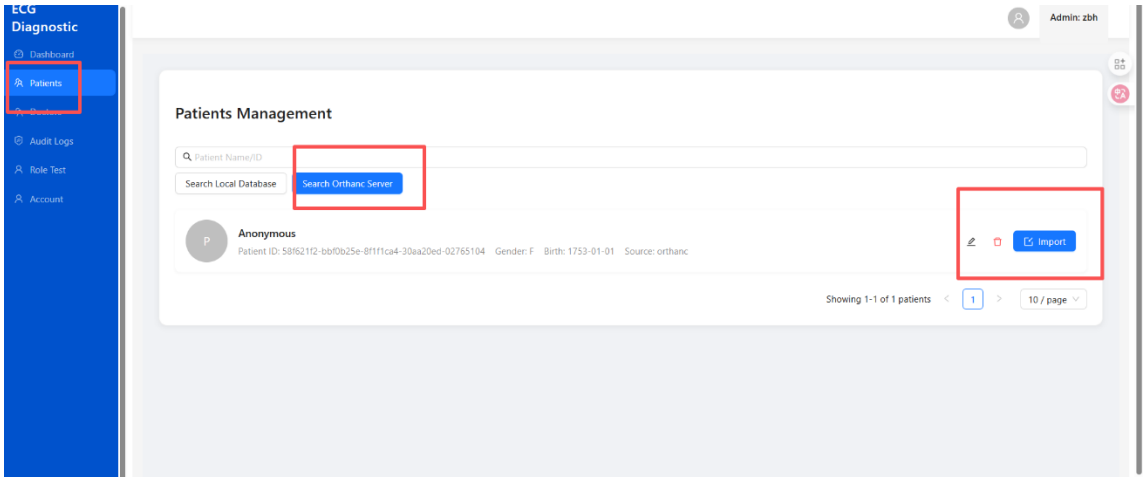


Usually, we shouldn't access the Orthanc server directly, so I add a test page to upload a test ECG file to Orthanc server, this test page url is :

http://localhost:8000/

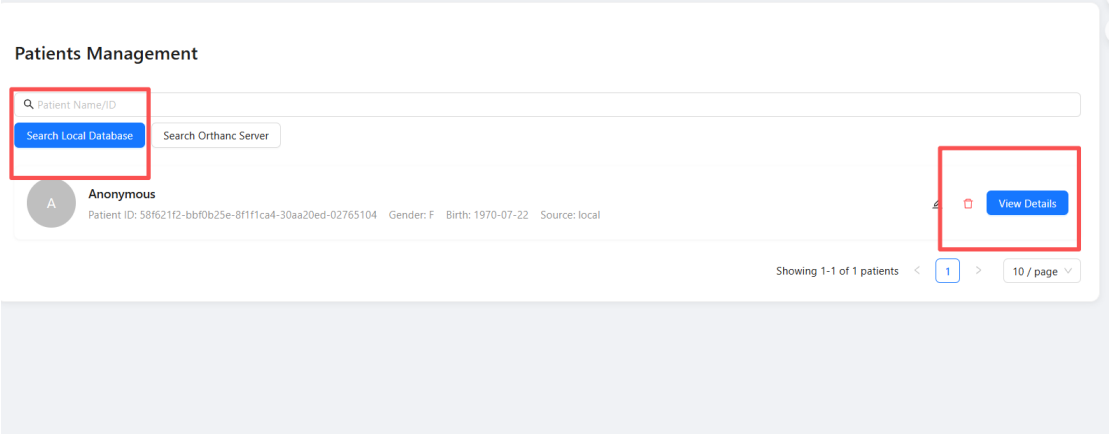


After we uploaded the ECG file to Orthanc server successfully. We could see it on the patient page.

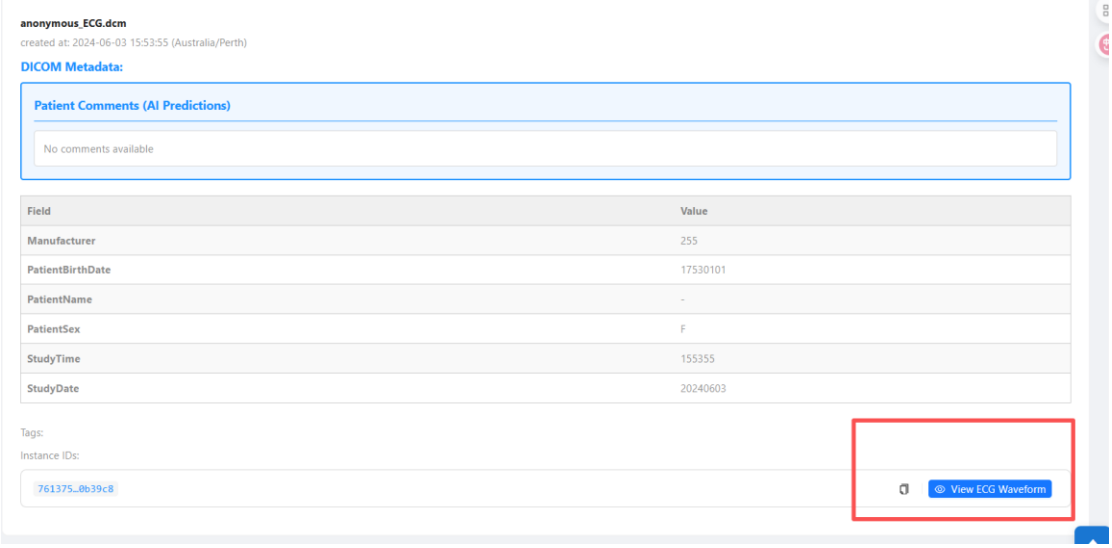


Step 3 diagnosis

We could import the patient into our database, and view the ecg file.



We can see this ECG file haven't been diagnosis, so we can check it's file first.



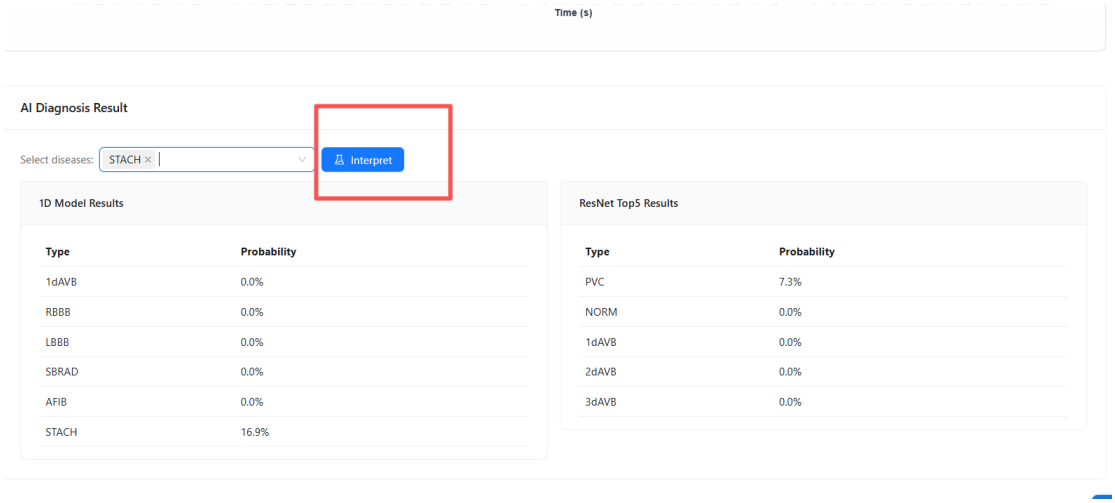


We could click the diagnosis button to get the diagnosis result.

Step 4 interpretation

We could the ai diagnosis result, and we could get the interpretation result.

First we select the disease we want to interpret.



We click the interpret button to get the interpretation result.

STACH ok

```
Obs for Sinus Tachycardia (943c2411-fd758e82-cff311d9-c85edc1c-64539d99) (Lead II):
- HR > 100 bpm. Avg: 99.8 bpm. RR intervals < 0.6s.
- For Sinus rhythms, P waves normal & precede QRS. Rhythm regular.
```

Metrics

average heart rate bpm: 99.845

mean rr interval s: 0.601

Images

sinus tachycardia analysis

